

1 Product Definition (Market-Real)

What DocuMind Is

A **secure SaaS platform** where organizations upload:

- Contracts
- HR policies
- SOPs
- Emails
- Technical documentation
- Scanned PDFs

and get:

A private, citation-accurate, AI assistant that answers questions ONLY from their own documents

No hallucinations. No mixing tenants. Full auditability.

Who pays?

Segment	Pain
Law firms	Searching contracts
SaaS companies	Onboarding & internal docs
HR departments	Policy interpretation
Consulting firms	Client document analysis
Compliance teams	Evidence & audits

Why companies will pay

Today they use:

- Ctrl+F
- SharePoint
- Email
- Notion

DocuMind replaces that with:

“Ask your company”

2 MVP Scope (4 Weeks, 5 Developers)

DocuMind MVP supports:

Capability	Included?
Multi-user SaaS	✓
Secure login	✓
File upload	✓
OCR	✓
RAG Q&A	✓
Citations	✓
Multi-agent	✓
Admin dashboard	✓
Usage tracking	✓
Billing-ready	API only (Stripe-ready)

What we **do NOT** build:

- Mobile apps

- Marketplace
- Advanced analytics
(Those are roadmap items)

System Architecture

Browser (React / Next.js)

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API Gateway

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Backend (FastAPI / ASP.NET / Node)

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| Auth | Billing | Users | Docs | Logs

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RAG Pipeline

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Vector DB (Pinecone / pgvector)

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LLM Orchestration (LangChain / Semantic Kernel)

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OpenAI / Claude APIs

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Langfuse Observability

Core AI Stack (Checklist-Compliant)

Requirement Implementation

Primary LLM OpenAI GPT-4.1

Fallback Claude 3.5

Embeddings text-embedding-3-large

Vector DB Pinecone or pgvector

Framework LangChain or Semantic Kernel

Observability Langfuse

Moderation OpenAI moderation API

OCR Azure AI Vision / Textract

5 RAG Pipeline (Gold Tier)

1. Document Ingestion

Supports:

- PDF
- DOCX
- TXT
- HTML
- Scanned images

Flow:

Upload → OCR → Text extraction → Metadata → Chunking → Embedding → Vector DB

2. Chunking Strategy

- Semantic chunking
- 800 tokens
- 15% overlap

- Header-aware

Stored metadata:

```
{  
  document_id,  
  page_number,  
  section,  
  uploaded_by,  
  date,  
  doc_type  
}
```

3. Retrieval

DocuMind uses **3 retrieval strategies**:

Type	Why
Semantic (vector)	Understand meaning
BM25 keyword	Exact terms
Hybrid fusion	Best of both

Plus:

- Reranking with cross-encoder
 - MMR for diversity
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4. Citations

Every answer returns:

Answer

+ List of source chunks

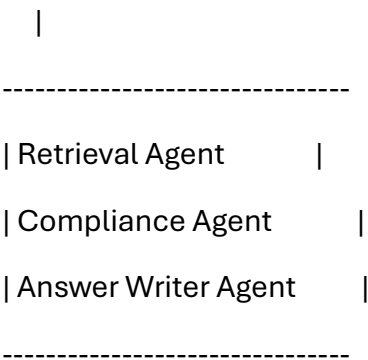
- + Page numbers
- + Downloadable original docs

This is what makes it **enterprise-grade**.

6 Multi-Agent Architecture

DocuMind uses **Hierarchical Agents** (Gold Tier).

Supervisor Agent



Roles

Agent	What it does
Supervisor	Plans the answer
Retrieval	Fetches documents
Compliance	Checks hallucinations
Writer	Produces final answer

They communicate via tool calls and shared memory.

7 Tools Available to Agents

Tool	Purpose
search_docs()	Vector + keyword search

Tool	Purpose
get_doc()	Fetch full document
check_compliance()	Validate grounding
log_trace()	Observability
get_user_role()	Access control

Multimodal Features

DocuMind implements **2 modalities** (Silver+):

Mode Use

Vision Read scanned contracts

Text Q&A

Optional roadmap:

- Voice Q&A
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Security (OWASP LLM Top 10)

Risk	How handled
Prompt injection	System prompt locked
Data leaks	Tenant-isolated vector DB
PII	Auto redaction
Tool abuse	Role-based access
Jailbreak	Input + output filters
Model DoS	Rate limiting
API leaks	Secrets manager

Zero-Trust Document Security

Every chunk stored with:

tenant_id

document_id

access_roles

LLM **never sees** data from other companies.

1 SaaS Features

Feature	Included
Multi-tenant	Yes
Roles (Admin/User)	Yes
API access	Yes
Usage dashboard	Yes
Token cost tracking	Yes
Stripe-ready	Yes

1 Why This is Real

DocuMind directly competes with:

- Chatbase (\$49–\$399/mo)
- Glean (enterprise)
- Hebbia (VC-backed)
- Humata

This is exactly what **AI startups are building right now.**

1 What students learn

They will implement:

- SaaS auth
- Vector DBs
- RAG
- Agents
- Multimodal OCR
- Cloud deployment
- Monitoring
- Security
- Cost control

This is **career-grade**, not academic.
